

Web Development

>>Introduction to HTML: Tags

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HTML - Definition

- **H**yper - is the opposite of linear
 - **T**ext – words / sentences / paragraphs
 - **M**ark-up – Marking the text
 - **L**anguage – It is a language
-
- It is used to define the structure and layout of a web page, how a page looks,
 - It is used to display content you read and see on the web.

How to write HTML?

- An HTML document is included of **one word commands enclosed between the less than (<) and greater than (>) sign**
- Determine how the page is handled
- Also called “TAGS”
- Tags **nearly always** come in pairs:
 - an opening tag `<html>`
 - a closing tag `</html>`

HTML Structure

- 5 Important HTML Tags you should know before coding an HTML document
 - **DOCTYPE**
 - **HTML**
 - **HEAD**
 - **TITLE**
 - **BODY**

Empty HTML Document

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>  
  
    </title>  
  </head>  
  <body>  
  
  </body>  
</html>
```

DOCTYPE Tag

- **DOC**ument **TYPE**
- Should be placed at the start of each web page, only one time
- Shows which standard the page complies with

HTML ver. 4.1

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"  
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

HTML ver. 5

```
<!DOCTYPE html>
```

HTML Tag

- Root Element / Tag
- Contains all tags that will be used in HTML document
- Necessary
- Attribute “lang”
 - define the language of the content within the document

```
<html lang="en">
```

```
</html>
```

HEAD Tag

- Provide information about the author, title and description of the page
- Also includes links to other scripting languages such as CSS and JavaScript
- meta tags (like keywords used by search engines)
- Includes the **<title>** tag
 - Includes a descriptive title of your web page

HEAD Tag

- meta tags

```
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
```

```
<meta name="description" content="Free Web tutorials" />
```

```
<meta name="keywords" content="HTML,CSS,XML,JavaScript" />
```

```
<meta name="author" content="John Doe" />
```

```
<meta http-equiv="refresh" content="5" />
```

```
<meta http-equiv="refresh" content="5" url=http://www.goole.com" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

Setting Cookies

- meta tags

```
<meta http-equiv="cookie" content=" userid=xyz; expires=Wednesday,  
14-Oct-25 23:59:59 GMT " />
```

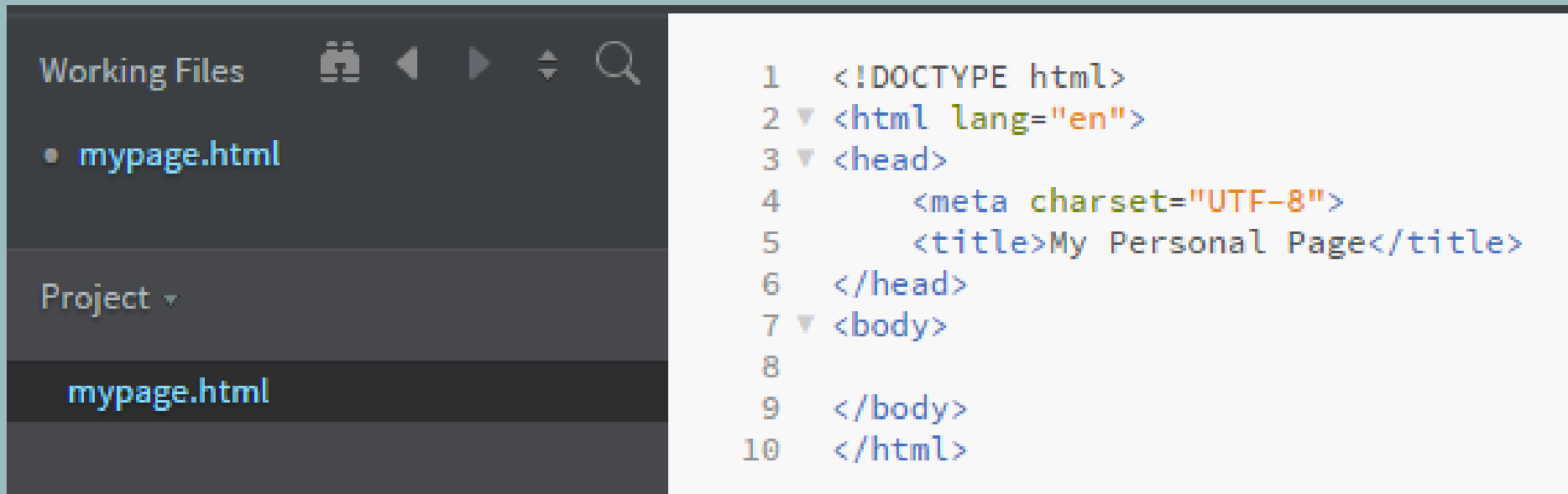
BODY Tag

- Contains information that will be displayed on the web
 - Text, Images, Videos, Ads, Links etc.
- ! Then tab → auto complete this lines
- Brackets editor
- Make new folder then right click → open this project in brackets
- Download extensions: Beautify,



TRY NOW

- Create a new html file and call it “**mypage.html**”
- Add the five required tags – doctype, html, head, title and body
- Add the title text as “**My Personal Page**”



The screenshot shows a code editor interface. On the left, a sidebar displays 'Working Files' with 'mypage.html' listed. Below it, the 'Project' section also shows 'mypage.html'. The main editor area on the right contains the following HTML code:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>My Personal Page</title>
6  </head>
7  <body>
8
9  </body>
10 </html>
```

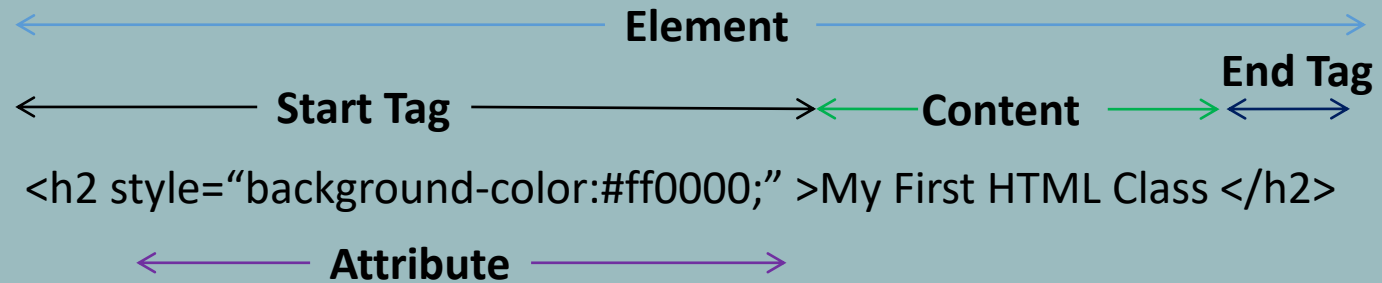
Text-Based Elements

in HTML

HTML TAGS

element = <opening tag> + content + <closing tag>

Tag Structure



Heading Tags

<h1>the biggest heading</h1>

<h2>the second biggest heading</h2>

.....

.....

.....

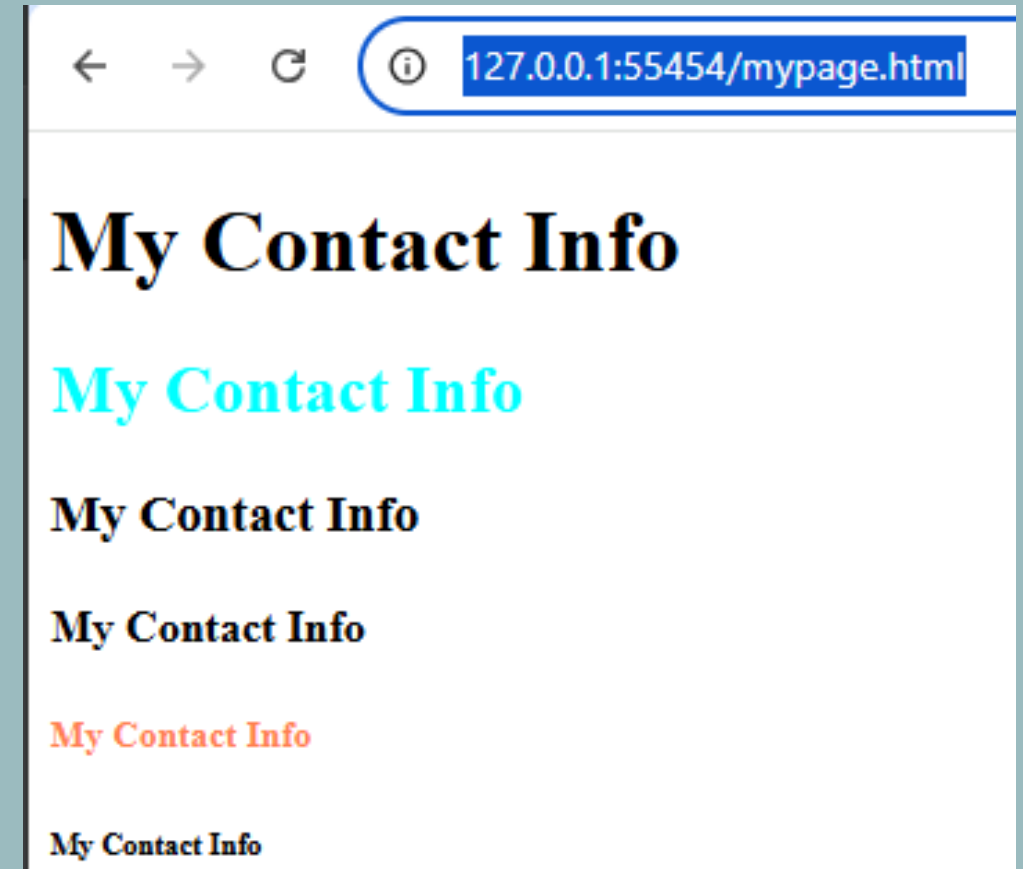
<h6>the smallest heading</h6>

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>SE are the champ</title>
6 </head>
7 <body>
8
9   <h1 style="color: crimson">We are the champ</h1>
10  |
11  |
12  <h2>We are the champ</h2>
13
14  <h3>We are the champ</h3>
15
16  <h6>We are the champ</h6>
17
18 </body>
19 </html>
```

TRY NOW

- Create a <h3> sized heading in the body
- Make the heading as "My Contact Info
(Just don't forget to close it)

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>My Personal Page</title>
6 </head>
7 <body>
8   <h1> My Contact Info </h1>
9   <h2 style="color:aqua"> My Contact Info </h2>
10  <h3> My Contact Info </h3>
11  <h4> My Contact Info </h4>
12  <h5 style="color: coral "> My Contact Info </h5>
13  <h6> My Contact Info </h6>
14 </body>
15 </html>
```



Paragraph Tag

Enclosed between the `<p>` and `</p>` tags

TRY NOW

- Create three paragraphs using `<p>` tags
- Fill them with content
 - Address
 - Telephone
 - Email Address

`<br>`: The Line Break element

- It produces a line break in text. It is useful for writing a *poem* or an *address*, where the division of lines is significant.
- This element has *no content* . It is called *empty* element.
- Empty elements also do **not** have an **end tag!**
- It can be written `<br/>` OR `<br>`

```

1  <!DOCTYPE html>
2  ▼ <html lang="en">
3  ▼ <head>
4      <meta charset="UTF-8">
5      <title>My Personal Page</title>
6  </head>
7  ▼ <body>
8      <h1> My Contact Info </h1>
9      <h2 style="color:aqua"> My Contact Info </h2>
10     <h3> My Contact Info </h3>
11     <h4> My Contact Info </h4>
12     <h5 style="color: coral "> My Contact Info </h5>
13     <h6> My Contact Info </h6>
14
15
16 ▼     <p>
17         the paragraph
18         <br> Heba Zaki <br> end of paragrapg
19
20     </p>
21 </body>
22 </html>

```

← → ↻ ⓘ 127.0.0.1:55454

My Contact Info

My Contact Info

My Contact Info

My Contact Info

My Contact Info

My Contact Info

the paragraph
Heba Zaki
end of paragrapg

Bold Tag

- To make text bold and place a strong importance on it
- Enclosed between the **** and **** tags

TRY NOW

- 
- For each of the `<p>` added before
 - Bold the titles
 - Address
 - Telephone
 - Email Address

Italics Tag

- To italicize text, thereby placing emphasis on it
- Enclosed between the `<em>` and `</em>` tags

TRY NOW

- 
- For each of the `<p>` added before
 - Italicize the content of each of them

- **`<hr>` added to make line separator**

Comments in HTML

- Content wrapped within a comment will not be displayed on the web page
- Comments help keep our files organized
- Comments become especially useful when there are multiple people working on the same files

HTML comments start with `<!--` and end with `-->`

EXAMPLE

`<!-- This is a comment in HTML -->`

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>My Personal Page</title>
6  </head>
7  <body>
8      <h1> My Contact Info </h1>
9      <h2 style="color:aqua"> My Contact Info </h2>
10     <h3> My Contact Info </h3>
11     <h4> My Contact Info </h4>
12     <h5 style="color: coral "> My Contact Info </h5>
13     <h6> My Contact Info </h6>
14
15
16     <p style="background-color: darkorange">
17         the paragraph
18         <br> Heba Zaki <br> end of paragrapg
19
20     </p>
21
22     <strong> Here is bold text</strong> <br>
23     <em> Address </em>
24
25
26     <!-- This is a comment in HTML -->
27
28
29 </body>
30 </html>

```

← → ↻ ⓘ 127.0.0.1:55454/mypage.html

My Contact Info

My Contact Info

My Contact Info

My Contact Info

My Contact Info

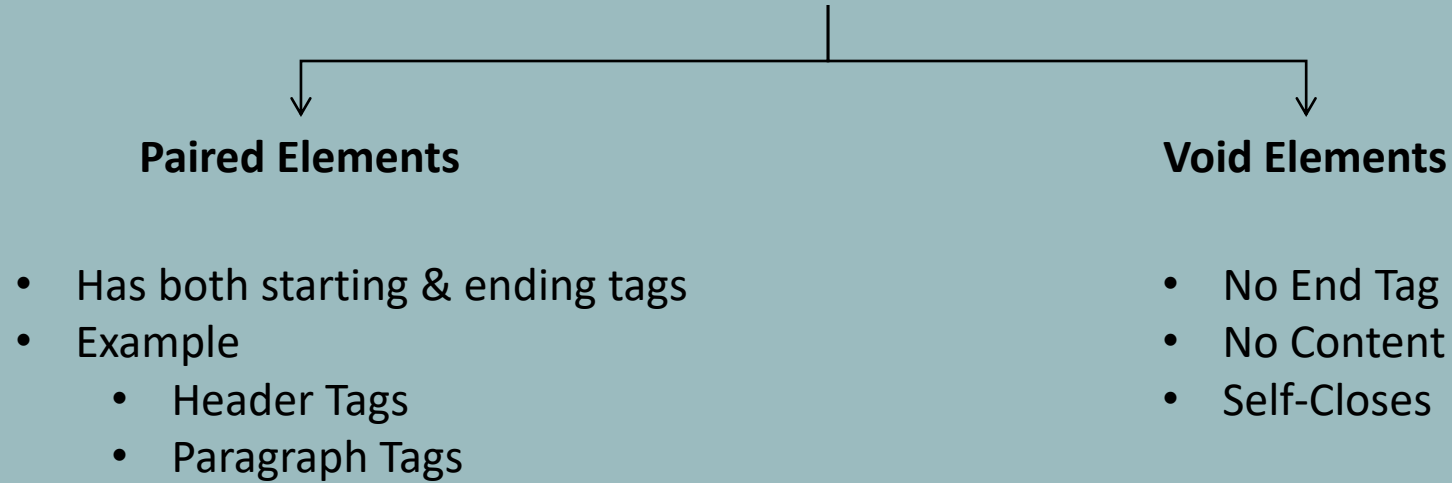
My Contact Info

the paragraph
Heba Zaki
end of paragrapg

Here is bold text

Address

HTML Elements



Summary

- HTML is used to give websites structure.
- We open HTML files using a browser, and the browser **renders**(shows us) the file.
- HTML files have a <head> and a<body> (just like you!)
- In the head, we have the <title>tags, and we use these to specify the webpage's name.
- How to make headings and paragraphs.
- How to bold and italicize text on the page

